

Mahdi Ghadamgahi

Machine Learning Engineer

Phone: +39 338 748 5390

Email:
ghadamgahi.smm@gmail.com |
Milan, Italy

LinkedIn:
[linkedin.com/in/ghadamgahi](https://www.linkedin.com/in/ghadamgahi)

PROFESSIONAL SUMMARY

Machine Learning Engineer with an MSc in Computer Science from Politecnico di Milano and 6+ years of research and industry experience designing and deploying production-grade AI systems. Specializes in computer vision, anomaly detection, and GenAI/LLM solutions for industrial and enterprise applications. Proven track record of delivering end-to-end ML pipelines — from data engineering and model development to production deployment — with measurable impact. Published researcher who bridges the gap between cutting-edge research and real-world engineering, communicating complex model behaviour clearly to both technical teams and business stakeholders.

TECHNICAL SKILLS

Python	LLM / GenAI	NLP / RAG	Model Deployment
Data Pipelines	PyTorch / TensorFlow	REST APIs / AWS, Azure, GCP	Docker / Git / CI-CD
Deep Learning	Explainable AI (XAI)	Anomaly Detection	Computer Vision
HuggingFace	LangChain / LangGraph	OpenCV / Scikit-learn	MLOps / Kubernetes

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

SCM Group | 2025 – 2026 | Rimini, Italy

- Developed an end-to-end synthetic data generation pipeline for automated defect detection in woodworking manufacturing, addressing critical data scarcity challenges in industrial computer vision.
- Implemented and benchmarked advanced deep learning approaches including GANs, CycleGANs, Stable Diffusion, and PatchCore for anomaly detection and localization; achieved 99.7% classification accuracy using ResNet18 with transfer learning on limited industrial datasets.
- Designed a robust anomaly detection system using PatchCore that maintained 98.8%+ AUROC with 20% ambiguous training data, enabling automated quality control for SCM's sizing process and reducing manual inspection costs.

AI Research Engineer

Iran's National Elites Foundation | 2021 – 2023 | Tehran, Iran

- Developed and trained fully connected neural networks to predict electricity distribution reliability indices across 11 feeders — achieving MSE of 0.0656 for failure rate prediction, representing a significant reduction in customer interruption costs compared to the conventional approach.
- Designed and implemented data preprocessing and fusion pipelines for heterogeneous IoT sensor streams, improving data quality and reliability for model training across Tehran's smart city infrastructure.
- Published 2 conference papers on AI-driven infrastructure maintenance (IEEE SGC2022, ICISE2022).

Machine Vision & Quality Control Engineer

IPCO – Irankhodro Powertrain Co. | 2020 – 2021 | Tehran, Iran

- Built and deployed computer vision models for real-time defect detection on factory conveyor belts, achieving 99% classification accuracy end-to-end from training to production inference.
- Applied explainability techniques (Grad-CAM) to make model decisions transparent to non-technical domain experts.
- Integrated real-time inference pipeline into production workflow, reducing manual inspection overhead by ~40% and improving throughput on the manufacturing line.

Machine Learning Research Engineer

Institute for Research in Fundamental Sciences (IPM) | 2018 – 2020 | Tehran, Iran

- Co-developed Iran's first MLaaS platform — designed REST API endpoints, model versioning, and scalable model-serving infrastructure, handling 10,000+ inference requests in production.
- Architected cloud-based model lifecycle management — an early form of modern MLOps and model-serving patterns.

- Benchmarked state-of-the-art algorithms across vision and NLP tasks; maintained close follow-up with AI research literature.

EDUCATION

MSc in Computer Science

Politecnico di Milano | 2023 – 2025

Thesis: Anomaly detection in industrial visual inspection using CNNs, CycleGAN, and PatchCore — supervised deep learning on manufacturing defect datasets.

BSc in Computer Engineering

Kharazmi University | 2017 – 2021

GPA: 4.0. Coursework in algorithms, data structures, machine learning, and software engineering.

LANGUAGES

- **English** — Fluent (working language)
- **Persian (Farsi)** — Native
- **Italian** — Basic

PUBLICATIONS

Improving Distribution Systems Maintenance Management, by Predicting Reliability Indices Using Fully Connected Neural Networks

12th Smart Grid Conference (SGC2022) | 2022 | DOI: [10.1109/SGC58052.2022.9998941](https://doi.org/10.1109/SGC58052.2022.9998941)

Tananaeifar M.H., Alimadady M., Ghadamgahi S.M.M., Ghafourian-Nasiri M., Fereidunian A.

Collaborative Design and Development of Integrated Preventive Maintenance Decision Support Software for Smart City Infrastructure based on Accumulated Data Analytics

8th International Conference on Industrial and Systems Engineering | 2022

Tananaeifar M.H., Ghafourian Nasiri M., Mohammadi P., Haghshenas M., Ghadamgahi S.M.M., Alimadady M., Fereidunian A.